

Shubhadeep Mandal

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EDUCATION

International Institute of Information Technology, Hyderabad <i>Telangana, India</i> <i>Master of Technology in Computer Science and Engineering</i>	July 2025 – Present CGPA: 9.12
Indian Institute of Information Technology, Ranchi <i>Ranchi, India</i> <i>Bachelor of Technology (Hons.) in Computer Science and Engineering</i>	Aug 2020 – Jun 2024 CGPA: 9.05

PROJECTS

BoardBuddy: Real-Time Collaborative Whiteboard <i>Next.js 14, Convex DB, Liveblocks, TypeScript</i> 🔗
– Developed a full-stack collaborative canvas supporting multi-user synchronization, freehand sketching, geometric shapes, and undo/redo. – Synchronized real-time state and user presence across sessions using Liveblocks WebSockets and Convex reactive database. – Secured organization-scoped workspaces and role permissions via Clerk authentication and responsive TailwindCSS/shadcn UI.
AI-SaaS: Multi-Modal Generation Platform <i>Next.js 13, TypeScript, Prisma, Stripe, PostgreSQL</i> 🔗
– Architected a multi-modal SaaS platform integrating OpenAI and Replicate APIs for conversational AI, code, audio, and video synthesis. – Engineered customer subscription billing, tiered usage quotas, and automated webhook event handlers using Stripe SDK and Clerk. – Optimized relational database queries and schema migrations with Prisma ORM deployed over managed PostgreSQL.
Real-Time Social Messaging Platform <i>Next.js, TypeScript, Pusher Channels, Prisma, MongoDB</i> 🔗
– Engineered instant 1-on-1 and group chat rooms with real-time WebSocket event broadcasting powered by Pusher Channels. – Implemented user online/offline presence tracking, active typing indicators, image uploads, and per-user message read receipts.
Peer-to-Peer Distributed File Sharing System <i>C++, POSIX Sockets, Multithreading, OpenSSL, Linux</i> 🔗
– Engineered a BitTorrent-style hybrid P2P system in C++, achieving an 8.5x transfer speedup on 1GB files (193s to 22.8s) across 5 concurrent peers by saturating network and disk I/O. – Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads. – Architected Primary-Backup tracker failover with 2s bounded failover, full state synchronization, and verified resilience via 48 unit tests under AddressSanitizer/UBSan.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, SQL, HTML/CSS, Bash.
Frontend Engineering: Next.js 14, React, TailwindCSS, shadcn/ui, Redux, HTML5 Canvas API, WebRTC.
Backend & Cloud: Node.js, Convex, Prisma ORM, PostgreSQL, MongoDB, REST APIs, Stripe SDK, Clerk.
Real-Time & Tools: WebSockets (Liveblocks, Pusher), Sockets, Linux/Unix, Git, Docker, Vercel.

ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

INTERESTS

- Web Novel and Manga Enthusiast.
- Chess and Badminton.